

Akhil Reddy

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PROFESSIONAL SUMMARY

Accomplished Senior Software Development Engineer with 5+ years of experience, recognized with multiple Kudos Awards for technical excellence in AL2 migration, CI/CD optimization, and dependency upgrades across enterprise systems. Expertise in Java, Kotlin, Spring Boot, Microservices, Kafka, and NoSQL/SQL databases with strong focus on system design, performance optimization, observability, and event-driven architectures. Adept at producing thoroughly tested, maintainable code while effectively collaborating with cross-functional teams.

CERTIFICATIONS

Oracle Cloud Infrastructure 2024 Generative AI professional

TECHNICAL SKILLS

- Languages** : Java, Kotlin, Python,, JavaScript, TypeScript, Go, Ruby, C/C++, PL/SQL, Shell Scripting
Frameworks : Spring Boot, Spring MVC, JPA, jQuery, React.js, Angular, Node.js, Flask, Django, ExpressJS, Bootstrap
DevOps and CI/CD : Docker, Kubernetes, Jenkins, Git, GitLab CI, GitHub, Azure DevOps, Maven, Gradle, Harness, Rancher
Cloud : AWS (EC2, S3, Lambda, EKS, CloudWatch, RDS, API Gateway, CodePipeline, Athena, IAM, ELB, RedShift), Azure , GCP
Messaging : Kafka, RabbitMQ, ActiveMQ, SQS
Database : Cassandra, MongoDB, DynamoDB, PostgreSQL, MySQL, Oracle, Microsoft SQL Server, Elasticsearch
Testing and Monitoring : JUnit, Selenium, JMeter, SoapUI, Splunk, CloudWatch, Postman, Swagger
Others : Microservices Architecture, System Design, Event-Driven Architecture, Agile/Scrum, SOLID Principles, Design Patterns

EXPERIENCE

- Senior Software Engineer** | Kroger February 2025 –Present
- **Video Monetization Feature:** Developed new video type for monetization on web platform, implementing premium placements and carousel functionality to enhance user engagement and revenue generation.
 - **Major Dependency Migration:** Led migration of 10+ microservices from Java 8 to Java 21, upgrading critical dependencies including Cassandra and Testcontainers, improving system performance and security
 - **OAuth 2.0 Implementation:** Implemented OAuth 2.0 authentication and authorization framework for secure API access, enhancing application security and user authentication workflows.
- Software Development Engineer** | Amazon Prime Video August 2022 – January 2025
- **Prime Traffic Insights Application:** Built traffic analysis web application using Java, Spring Boot, Python, and React.js, optimizing traffic distribution for 13M+ viewers during high-demand events like Thursday Night Football.
 - **URL Service Re-architecture:** Re-designed URL Service into modular APIs using Java and Spring Boot, achieving 99.99% availability for multi-region CDN routing operations.
 - **Amazon Linux 2 Migration:** Led migration of Prime Video systems to AL2, reducing infrastructure costs by 25% while improving performance and security for global streaming services.
 - **CI/CD Pipeline Automation:** Automated deployment workflows using AWS CodePipeline and Ruby SDKs, implementing automated rollbacks, CloudWatch alarms, and metrics monitoring, saving 20 hours per week and increasing deployment frequency by 40%.
- Software Engineer** | Subex January 2019 – August 2021
- **Payment Workflow Optimization:** Designed batch processing systems with scheduled jobs, reducing transaction costs and saving \$250K annually.
 - **Microservices Architecture:** Built scalable microservices using Spring Boot, Hibernate, and Kafka, cutting deployment time by 30% and enhancing real-time communication.
 - **Full-Stack Development:** Developed responsive user interfaces using Angular, HTML5, and CSS3, creating reusable components and services to consume REST APIs for seamless frontend-backend integration.

EDUCATION

University of Cincinnati **MS in Information Technology** | GPA : 3.93 August 2021 – August 2022

PROJECTS

- RAG-Based Document Query Chatbot** | Python (FastAPI, LangChain), React.js, ChromaDB, Pinecone, Ollama , OpenAI API
- Built intelligent chatbot using RAG architecture with FastAPI, ChromaDB vector database, and Sentence Transformers for semantic document retrieval from PDF manuals.
 - Integrated Ollama (Phi3) for local LLM inference with prompt engineering, achieving accurate context-aware responses while processing 500+ text chunks with embeddings.
 - Developed full-stack solution with React frontend and Python backend, implementing RESTful APIs for real-time chat and vector similarity search functionality.
- Gesture Based Smart Shopping Using Deep Learning** | Python, TensorFlow/PyTorch, CNN, RNN, U-Net, OpenCV
- Created accessible shopping application for kiosk machines using deep learning frameworks (TensorFlow/PyTorch), implementing CNN, RNN, VGG-16, and U-Net architectures for real-time gesture recognition and navigation.
 - Performed channel boosting for the CNN model to achieve 94% accuracy.